Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____ Calculator: yes

Learning goal: I can compare two data sets using their displays. (circle your level after scoring)

4	I can make and justify a claim about two populations using their data displays.
3	I can compare two data sets using measures of center and spread from their displays.
2	I can read values from a box plot or histogram.
1	I need help to start this benchmark.

1. Route A: 30, 28, 26, 30, 26
Route B: 21, 20, 24, 20, 25

The lists show the times, in minutes, of five trips on each of two bus routes. Which conclusion do the means of the two data sets support?

A. Trips on Route A typically took about 6 more minutes than trips on Route B.

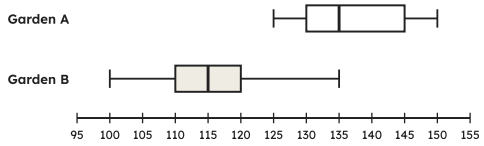
B. Trips on Route A typically took about 30 more minutes than trips on Route B.

C. Trips on Route A typically took about 5 more minutes than trips on Route B.

D. Trips on Route B typically took about 6 more minutes than trips on Route A.

2. The box plots show the heights, in centimeters, of the sunflowers grown in two gardens. The two gardens have the same number of sunflowers.

A sunflower from one of the two gardens is 132 centimeters tall. Which garden did the sunflower more likely come from, and why?



A. Garden A, because 132 is in the middle half for Garden A but in the top fourth for Garden B

B. Garden B, because the median of the data for Garden B is closer to 132 than the median for Garden A

C. Garden B, because 132 is between the least and the greatest values of the data for Garden B

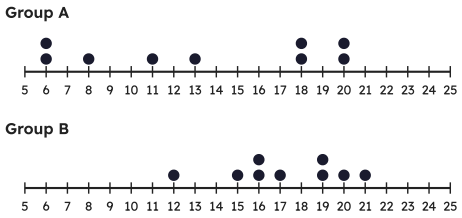
D. Either one is equally likely, because 132 is within the range of the data for both gardens

Turn over →

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____

3. The line plots show the numbers of pages read last night by the 9 students in each of two reading groups.

Which conclusion do the medians of the two groups support?



A. Students in Group B typically read about 1 more pages than students in Group A.

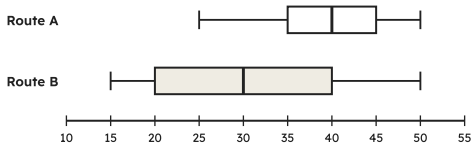
B. Students in Group B typically read about 6 more pages than students in Group A.

C. Students in Group A typically read about 4 more pages than students in Group B.

D. Students in Group B typically read about 4 more pages than students in Group A.

4. The box plots show the times, in minutes, of trips on two bus routes.

Which conclusion about the two routes is supported by the box plots?



A. Trips on Route A typically took longer, and the trip times on Route B varied more.

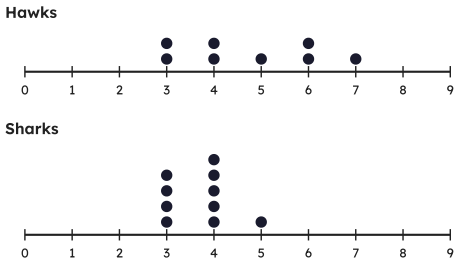
B. Trips on Route A typically took longer, and the trip times on Route A varied more.

C. Trips on the two routes typically took the same amount of time, and the trip times on Route A varied more.

D. Trips on Route B typically took longer, and the trip times on Route B varied more.

5. The line plots show the numbers of goals two soccer teams scored in their games.

Use the mean and the range of each data set. Which conclusion is supported by the data?



A. The Sharks typically scored more goals, and the Hawks' numbers of goals varied more.

B. The Sharks typically scored more goals, and the Sharks' numbers of goals varied more.

C. The Hawks typically scored more goals, and the Sharks' numbers of goals varied more.

D. The Hawks typically scored more goals, and the Hawks' numbers of goals varied more.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.2 · MA.7.DP.1.2 · Compare two data sets

Unit 13: Numerical and Categorical Data. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. Trips on Route A typically took about 6 more minutes than trips on Route B.**

Route A: $\text{mean} = 140 \div 5 = 28$.
Route B: $\text{mean} = 110 \div 5 = 22$.
$28 - 22 = 6$, so: Trips on Route A typically took about 6 more minutes than trips on Route B.

2. Regular **A. Garden A, because 132 is in the middle half for Garden A but in the top fourth for Garden B**

Garden A: the box runs from 130 to 145, so 132 is among the middle half of its values.
Garden B: the upper quartile is 120, so only about one-fourth of its values are greater than 120, and they are spread out from 120 to 135.
The gardens are the same size, and far more values of Garden A lie near 132, so the sunflower is more likely from Garden A.

3. Regular **D. Students in Group B typically read about 4 more pages than students in Group A.**

Each plot has 9 dots, so the median is the 5th dot counting from either end.
Group A: in order, 6, 6, 8, 11, 13, 18, 18, 20, 20; the median is 13.
Group B: in order, 12, 15, 16, 16, 17, 19, 19, 20, 21; the median is 17.
$17 - 13 = 4$, so: Students in Group B typically read about 4 more pages than students in Group A.

4. Regular **A. Trips on Route A typically took longer, and the trip times on Route B varied more.**

Route A: median 40 minutes; interquartile range $45 - 35 = 10$; range $50 - 25 = 25$.
Route B: median 30 minutes; interquartile range $40 - 20 = 20$; range $50 - 15 = 35$.
The median shows the typical value, and the interquartile range and the range show how much the values vary. Trips on Route A typically took longer, and the trip times on Route B varied more.

5. Rigor **D. The Hawks typically scored more goals, and the Hawks' numbers of goals varied more.**

Hawks: $\text{mean} = 38 \div 8 = 4.75$; $\text{range} = 7 - 3 = 4$.
Sharks: $\text{mean} = 37 \div 10 = 3.7$; $\text{range} = 5 - 3 = 2$.
The greater mean shows the greater typical value, and the greater range shows more variation. The Hawks typically scored more goals, and the Hawks' numbers of goals varied more.

Re-check strip (3 items, C-DP.1.2-R)

1. Regular **C. Team A typically scored about 5 more points per game than Team B.**

Team A: $\text{mean} = 250 \div 5 = 50$.
Team B: $\text{mean} = 225 \div 5 = 45$.
$50 - 45 = 5$, so: Team A typically scored about 5 more points per game than Team B.

2. Regular **A. Class A, because 42 is in the middle half for Class A but in the top fourth for Class B**

Class A: the box runs from 35 to 50, so 42 is among the middle half of its values.
Class B: the upper quartile is 25, so only about one-fourth of its values are greater than 25, and they are spread out from 25 to 45.
The classes are the same size, and far more values of Class A lie near 42, so the student is more likely from Class A.

3. Regular **D. Students in the two classes typically scored the same, and the scores in Class A varied more.**

Class A: median 25 points; interquartile range $35 - 15 = 20$; range $40 - 10 = 30$.
Class B: median 25 points; interquartile range $30 - 20 = 10$; range $35 - 10 = 25$.
The median shows the typical value, and the interquartile range and the range show how much the values vary. Students in the two classes typically scored the same, and the scores in Class A varied more.

Comparison strip (2 items, C-DP.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. Students in Class B typically did about 6 more push-ups than students in Class A.**

Each plot has 9 dots, so the median is the 5th dot counting from either end.
Class A: in order, 5, 5, 7, 9, 12, 13, 13, 15, 19; the median is 12.
Class B: in order, 12, 13, 15, 16, 18, 19, 19, 23, 23; the median is 18.
$18 - 12 = 6$, so: Students in Class B typically did about 6 more push-ups than students in Class A.

2. Regular **C. Garden B, because 139 is in the middle half for Garden B but in the top fourth for Garden A**

Garden B: the box runs from 130 to 145, so 139 is among the middle half of its values.
Garden A: the upper quartile is 125, so only about one-fourth of its values are greater than 125, and they are spread out from 125 to 140.
The gardens are the same size, and far more values of Garden B lie near 139, so the sunflower is more likely from Garden B.

Reteach plan — C-DP.1.2

MA.7.DP.1.2 · Compare two data sets

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.2-R

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____ Calculator: yes

1. Team A: 48, 52, 55, 48, 47
Team B: 49, 43, 45, 45, 43

The lists show the points scored in five games by each of two basketball teams. Which conclusion do the means of the two data sets support?

A. Team A typically scored about 25 more points per game than Team B.

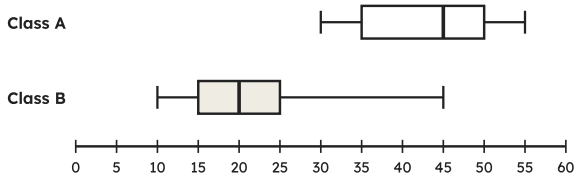
B. Team A typically scored about 6 more points per game than Team B.

C. Team A typically scored about 5 more points per game than Team B.

D. Team B typically scored about 5 more points per game than Team A.

2. The box plots show the numbers of books read over the summer by the students in two classes. The two classes have the same number of students.

A student from one of the two classes read 42 books over the summer. Which class is the student more likely in, and why?



A. Class A, because 42 is in the middle half for Class A but in the top fourth for Class B

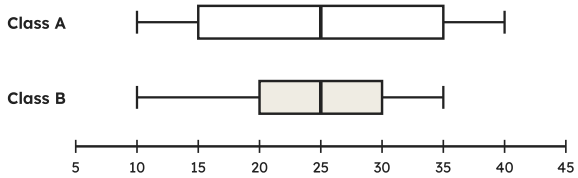
B. Class B, because 42 is between the least and the greatest values of the data for Class B

C. Class B, because the median of the data for Class B is closer to 42 than the median for Class A

D. Either one is equally likely, because 42 is within the range of the data for both classes

3. The box plots show the quiz scores of the students in two classes.

Which conclusion about the two classes is supported by the box plots?



A. Students in Class A typically scored higher, and the scores in Class A varied more.

B. Students in Class B typically scored higher, and the scores in Class A varied more.

C. Students in Class A typically scored higher, and the scores in Class B varied more.

D. Students in the two classes typically scored the same, and the scores in Class A varied more.

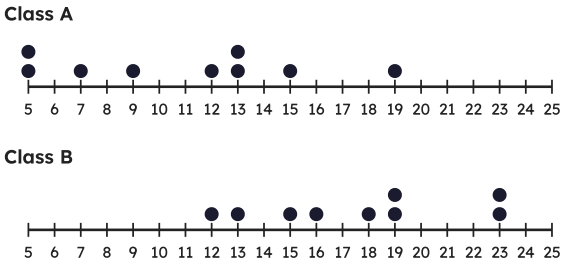
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.2-X

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____ Calculator: yes

1. The line plots show the numbers of push-ups done in one minute by the 9 students in each of two classes.

Which conclusion do the medians of the two groups support?



A. Students in Class B typically did about 6 more push-ups than students in Class A.

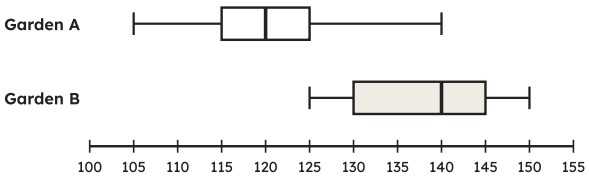
B. Students in Class A typically did about 6 more push-ups than students in Class B.

C. Students in Class B typically did about 7 more push-ups than students in Class A.

D. Students in Class B typically did about 4 more push-ups than students in Class A.

2. The box plots show the heights, in centimeters, of the sunflowers grown in two gardens. The two gardens have the same number of sunflowers.

A sunflower from one of the two gardens is 139 centimeters tall. Which garden did the sunflower more likely come from, and why?



A. Garden A, because 139 is between the least and the greatest values of the data for Garden A

B. Either one is equally likely, because 139 is within the range of the data for both gardens

C. Garden B, because 139 is in the middle half for Garden B but in the top fourth for Garden A

D. Garden A, because the median of the data for Garden A is closer to 139 than the median for Garden B

Score: ____ / 2 → copy to your progress strip